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MILITARY SPECIFICATION

RELAY, CURRENT, TYPE A-3 (BATTERY AND STARTER CIRCUIT)

1. SCOPE

1.1 This specification covers one type of 12V dc aircraft battery and starter circuit control relay, designated Type A-3.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND PUBLICATIONS

2.1 The following specifications, standards, and publication, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-B-601	Boxes, Wood-Cleated-Plywood, For Domestic Shipment
NN-B-621	Boxes, Wood, Nailed and Lock- Corner
NN-B-631	Boxes; Wood, Wirebound (For Domestic Shipment)
LLL-B-631	Boxes; Fiber Corrugated (For Domestic Shipment)
LLL-B-636	Boxes; Fiber, Solid, (For Domestic Shipment)

MILITARY

MIL-P-14	Plastic-Materials, Molding, and Plastic-Parts, Molded; Thermosetting
MIL-P-79	Plastic-Materials, Laminated, Thermosetting, Rods and Tubes
JAN-P-105	Packaging and Packing for Overseas Shipment, Boxes, Wood, Cleated, Plywood
JAN-P-106	Boxes; Wood, Nailed, (Military Type)
MIL-P-116	Preservation, Methods of
JAN-P-120	Packaging and Packing for Overseas Shipment, Cartons, Folding Paperboard
JAN-P-125	Packaging and Packing for Overseas Shipment- Barrier-Materials, Waterproof, Flexible
JAN-P-133	Packaging and Packing for Overseas Shipment- Boxes, Set-Up Paperboard
MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier-Material

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MIL-B-4229	Boxes; Paperboard, Metal Edge
MIL-D-5028	Drawings and Data Lists; Preparation of (For Engines, Accessories and Other Auxiliary Equipment)
MIL-S-7742	Screw Threads; Standard, Aeronautical

STANDARDS

MILITARY

MIL-STD-129	Marking of Shipments
MIL-STD-120	Identification of U. S. Military Property

PUBLICATION

AIR FORCE-NAVY AERONAUTICAL BULLETIN

143	Specifications and Standards; Use of

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by thecontractingofficer.)

3. REQUIREMENTS

3.1 Materials:

3.1.1 Specifications and standards. - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, *shall be selected in accordance with Bulletin No.* 143, except as provided in the following paragraph:

3.1.1.1 Standard parts. - AN or MIL Standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL Standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.1.1.2 All exposed steel parts shall be corrosion resistant steel or shall be suitably plated to resist corrosion.

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3.1.2 Protective treatment. - When materials are used in the construction of the relay that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.2 Design and construction. - The relay shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.2.1 The relay shall be constructed so that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases.

3.2.2 The relay shall be suitable for operation at temperatures between -55 and +71°C. The relay shall not be damaged when exposed to temperatures of 93.3°C for a minimum period of 48 hours.

3.2.3 The relay shall conform to the dimensions shown on Figure 1.

3.2.4 The construction of the relay shall insure proper operation when mounted in any position, and the contacts and holding coil shall be totally inclosed for mechanical and dust protection.

3.2.5 The design of the relay shall permit disassembly and reassembly for the purpose of repair or replacement of parts without special tools.

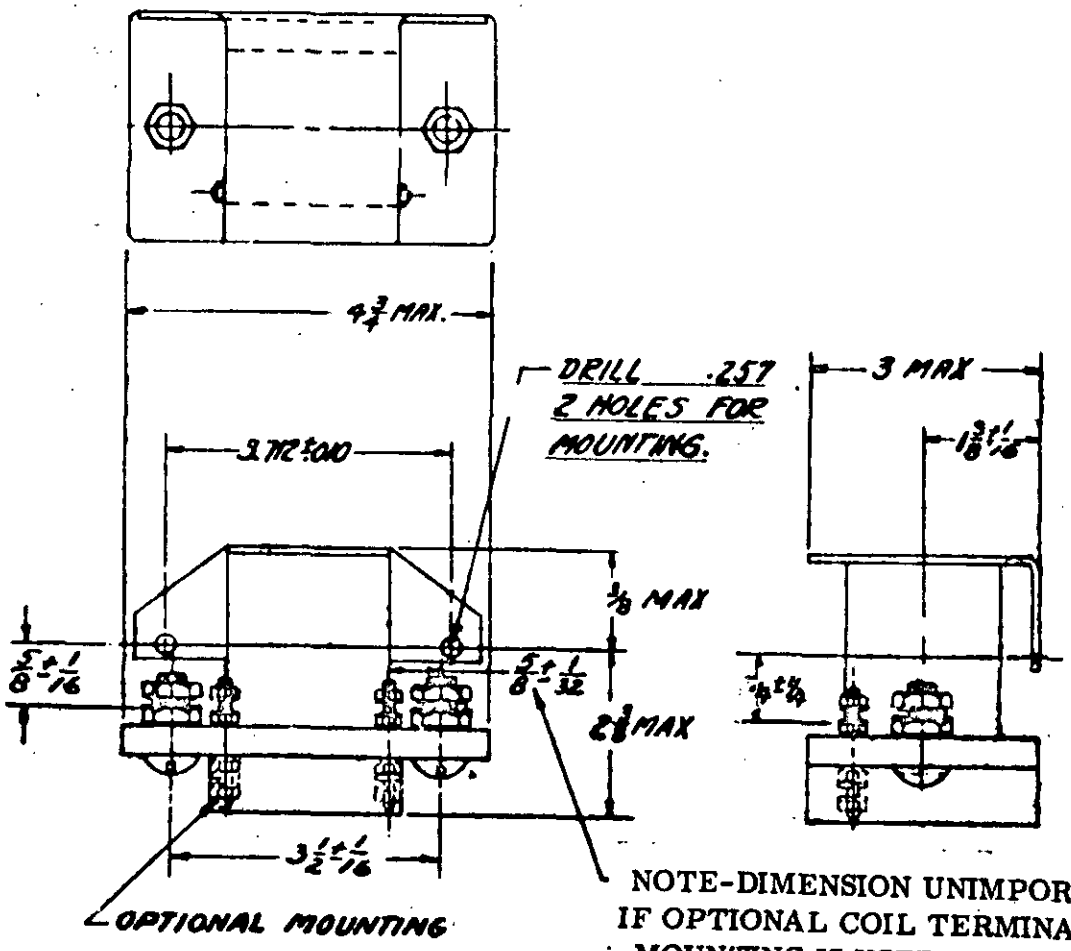
3.2.6 The relay mounting bracket shall be an integral part of the relay housing or shall be securely attached thereto in a manner to prevent any movement between the relay and mounting bracket in service use.

3.2.7 Operating voltage. - The relay shall be designed so that it will withstand continuous operation on a 14.5V dc source with a maximum temperature rise of 70°C on the coil at an ambient temperature of 25°C.

3.2.8 Closing voltage. - The relay shall be designed so that the main switch contacts close firmly with a maximum direct current potential of 9.0V applied to the terminals of the holding coil, and shall hold closed on 3.5V when the closing voltage applied to the holding coil is reduced from 9.0 to 3.5V after one hour of continuous operation on 9.0V.

3.2.9 Relay holding coil current consumption. - The current consumed by the relay holding coil when operated on 14.5V dc shall not exceed 1.0 nor more than 2.0 amp when operated at the same voltage -54°C.

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DIMENSIONS IN INCHES.

FIGURE 1

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3.2.10 Holding coil winding. - The relay holding coil shall be evenly wound, properly insulated, and taped and impregnated by the vacuum pressure process with a moisture-proof and heat-resistant protective material.

3.2.11 Holding coil heads. - The relay holding coil heads, if used, shall be of phenolic material conforming to Specification MIL-P-14, or MIL-P-79.

3.2.12 Terminal base. - The terminal base shall be of phenolic material conforming to Specification MIL-P-14 or MIL-P-79.

3.3 Performance. - Unless otherwise specified, the relay shall be capable of operating satisfactorily under the following conditions:

3.3.1 Temperature. - Temperature ranging from -54 to +71°C.

3.3.2 Vibration. - Vibration incident to service use.

3.3.3 Acceleration and shock. - Acceleration and shock forces as anticipated.

3.4 Terminal posts:

3.4.1 Both leads of the relay holding coil shall be brought out on one end of the relay to terminal post.

3.4.2 The terminal posts shall be threaded with No. 6-32, NC-2 threads, in accordance with Specification MIL-S-7742, and shall be provided with the proper nuts and lock washers for attaching cable terminals.

3.5 Relay terminals:

3.5.1 Both insulated relay terminal posts shall be brought out on the same end and properly spaced from those of the holding coil.

3.5.2 Both insulated relay terminal posts shall be threaded with 3/8-24, NF-2 threads in accordance with Specification MIL-S-7742, and shall be provided with the proper nuts and lock washers for attaching cable terminals. The terminal posts shall be of sufficient length to provide a minimum of 3/8 inch free threads beyond the nuts and washers when in the secured position.

3.6 Capacity. - The main contacts of the current relay shall be capable of making and breaking 800 amp dc at 12V for 50 operations at the rate of one second "on" and 4 seconds "off" without welding of the contacts.

3.7 Main contact drop. - With a potential of 12V on the relay holding coil and 200 amp dc flowing through the main contacts, the voltage drop across the main contacts

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shall not exceed 0.10V after making and breaking 200 amp at 12V for 50,000 operations at a rate of 60 operations per minute.

3.8 Insulation. - The current relay shall be insulated so that it will withstand between the terminals and exposed metal parts, an alternating current of 1000V rms, 60 cycles for one minute.

3.9 Interchangeability. - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.10 Weight. - The weight of the relay shall not be more than 2.35 pounds.

3.11 Identification of product. - Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.12 Workmanship:

3.12.1 General. - The relay, including all parts and accessories, shall be fabricated in a workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection tests:

4.1.1 Individual tests. - Each relay shall be subjected to the following tests:

4.1.1.1 Examination of product. - The relay shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.1.1.2 Each relay shall be subjected to a potential of 1000V rms at commercial frequency between each main contact and holding coil terminal and between all terminals and other exposed metal parts for 60 seconds. The relay shall withstand this test without failure.

4.1.2 When quality control tests are specified on anumber of items that are selected from a production run and one or more of this number fails to meet the specified tests, acceptance of all items still on hand and subsequent production will be with held until the extent and cause of failure is determined. For operational reasons, individual tests maybe continued pending investigation of a quality control test failure. However, final acceptance of the items on hand and subsequent production is contingent upon the inspector's decision regarding the overall conformance of

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the product to specification requirements. When corrective action has been accomplished, all necessary tests shall be repeated. If investigation indicates that the defects may exist in items previously accepted, full particulars concerning the defects, including recommendations for correction, will be furnished to the contracting officer.

4.1.2.1 Sampling tests. - Relays shall be selected at random as specified below and subjected to the following tests:

Number of relays selected	Quantity on order
All	10 or less
5	11 to 100
8	101 to 300
1 additional for each 100 or fraction thereof above 300	

4.1.2.2 Unless otherwise specified, the current relays shall be tested at temperatures between 20 and 30°C.

4.1.2.3 Each relay shall be subjected to tests to determine if the main contacts close with a direct current potential of 9.0V applied to the holding coil terminals and to determine if the main contacts remain closed when the voltage across the holding coil is reduced from 9.0 to 3.5V. With 12V applied to the coils terminals, the coil current shall not be greater than one amp.

4.1.2.4 One relay shall be exposed to a temperature of -55°C for a period of 48 hours. At the end of this exposure period and while still at this temperature, the relay shall be tested to determine that the contacts close when a potential of 9V dc is applied to the holding coil and to determine that the contacts remain closed when the voltage is reduced from 9 to 3.5V. With 14-1/2V dc applied to the coil terminals the coil current shall not be greater than 2 amp.

4.1.2.5 The relay shall then be subjected to a temperature of 70°C for a period of 16 hours at the conclusion of which, and while still at this temperature, the relay shall be tested to determine that the contacts close when a potential of 9V is applied to the holding coil and to determine that the contacts remain closed when the coil voltage is reduced to 3-1/2V.

4.1.2.6 One relay shall be subjected to a temperature of 93.3°C for a minimum period of 48 hours and shall, after having attained room temperature, be subjected to 50,000 cycles of operation making and breaking a circuit of 200 amp at 14-1/2V dc. During this test, a potential of 12V dc shall be applied across the coil terminals for a period of 1/2 second during each cycle. The rate of cycling shall be 60 per minute. The relay shall operate without welding of the contacts. At the conclusion of this test, the voltage drop across the main contacts while carrying 200 amp shall be no greater than 0.10 with 12V dc applied to the holding coil.

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4.1.2.6.1 After meeting the tests specified in 4.1.2.6, the relay shall be operated for 50 cycles of make and break at a rate of one second "on" and 4 seconds "off" at 400 percent of rated current at rated voltage non-inductive load without welding of the contacts.

4.1.2.7 Coil temperature rise. - One relay shall be tested to determine that the coil temperature rise is no greater than 70°C. For the purpose of this test, the relay shall be subjected to a resistive load of 200 amp through the contacts which are held closed for a period of 3 hours with 14.5V dc applied to the holding coil. Immediately following the 3 hour period, the resistance of the coil shall be measured and the value of this resistance shall be used in conjunction with the resistance of the relay to room temperature to determine the coil temperature rise by the resistance method.

4.1.2.8 Vibration. - One relay shall be subjected to vibration tests. Vibration shall consist of simple harmonic motion having an amplitude of 0.03 inch (maximum total excursion of 0.06 inch) and a frequency varied uniformly between the approximate limits of 10 to 55 cps. The entire cycle of frequencies shall be accomplished in approximately one minute. The relay shall be mounted in the position most likely to cause malfunctioning and vibrated continuously for a period of 3 hours. If mounting in more than one position is considered likely to cause malfunctioning, the relay shall be vibrated continuously for 3 hours in each likely position. A test lamp or suitable indicating device shall be connected across the main contacts while the relay is undergoing this test to determine that the vibration does not close the contacts.

4.1.2.9 Acceleration. - One relay shall be subjected to accleration forces of 10g. The time required to attain the 10g acceleration shall be not more than 2 seconds. The relay shall be mounted in the position most likely to cause malfunctioning and shall be accelerated for a minimum of one minute without causing the main contacts to close. A test lamp or a suitable indicating device shall be connected across the main contact terminals while the relay is undergoing this test to determine that the acceleration does not result in closing the contacts.

4.2 All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with the specification, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 General. - The requirements specified herein apply only to direct purchases by, or direct shipments to the Government.

5.2 Unit packaging. - The relays shall be preserved in accordance with Specification MIL-P-116, Method IIc and shall be indvidually packaged in containers con-

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forming to Specification JAN-P-120, JAN-P-133, or MIL-B-4229.

5.3 Exterior packing:

5.3.1 Domestic. - A uniform number of unit packages shall be packed in exterior type containers conforming to Specification NN-B-601, NN-B-621, NN-B-631, LLL-B-631, or LLL-B-636. When fiberboard exterior containers are used, such containers shall be fabricated from fiberboard having a Mullen test of 275 pounds or more.

5.3.2 Export. - Unless otherwise specified, a uniform number of domestic packages shall be packed in a container conforming to Specification JAN-P-105 or JAN-P-106. Plywood, if used shall be Type B, Condition I. Each container shall be lined with a waterproof barrier in accordance with Specification JAN-P-125 and sealed with adhesive conforming to Specification MIL-A-140.

5.4 Marking of shipments. - Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows:

Relay, Current (Battery and Starter Circuit)
Type A-3
Specification MIL-R-9213
Mfr's. Part No. *

* Applicable data shall be entered by the contractor.

6. NOTES

6.1 Use. - The Type A-3 current relay covered by this specification is intended for use in airplanes as a means of remotely controlling the making and breaking of the battery and starter circuit in close proximity to the battery and starter.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government Procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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